LiDAR | Complete Illustrated Research Report

GEMSTAR Task 23: OSI decoding, scanning geometry, controlled calibration, weather availability, MATLAB and CPU scoring timing.

Evidence source: uploaded Task_23_Final_Research.zip. Figures and tables come from archived assets. This document explains established findings separately from assumptions and future validation.

Executive research conclusions

LiDAR work includes raw OSI inspection, ray/range reconstruction, source-driven geometry correction, vehicle-frame transformation, ground-truth actor-ID exploration, controlled 2 m cube displacement, Clear/Fog100/Fog20 return-availability analysis, MATLAB cross-checks, and preliminary Python scoring timings. Most experiments use a static simulation arena. Positive returns are availability, not detection correctness.

Controlled weather	Frames audited	Returns / slots per frame	Availability
Clear	13,819	5,090 / 5,271	96.5661%
Fog MOR 100	12,386	5,090 / 5,271	96.5661%
Fog MOR 20	14,597	0 / 5,271	0%

Total reported scored frames: 40,802. Identical within-scenario return patterns limit independent sampling claims.

How each main LiDAR subtask contributed

23.2 - 23.4A | OSI structure and LiDAR input integrity

Inspect OSI sensor fields, reflection arrays, valid TOF/signal fields, and frame/ray structure. This prevents incorrect data types or missing-reflection fields from entering later algorithms.

23.4B-4D | Ray geometry and source correction

Calculate range from simulated reflections, construct range images/3D point clouds, inspect ray ordering and rejected rows, then correct reconstruction using actual source-defined scan geometry. MATLAB was used to cross-check ranges and range images.

23.4E | Vehicle and ground-truth coordinate analysis

Audit sensor mount, host pose, source-defined sign conventions, actor ID and box geometry. These are exploratory; object correspondences and absolute localization accuracy were not fully established.

23.8 | Controlled LiDAR displacement

A 3 m cube moved 2 m. The final report gives 765 matching rays among 1,071 candidate rays, median matched range change 2.013944 m. These are selected geometric candidates, not a target-level true-positive metric.

23.10-11A/B/C | Weather setup, integrity, preprocessing

Configure/verify controlled fog, audit OSI integrity and timestamps, produce and validate processed NPZ masks. These establish data readiness and prevent malformed frames being treated as physical failures.

23.11D/E | Clear baseline and three-weather recording

Compare Clear, Fog MOR 100 m, and Fog MOR 20 m in a static geometric scene; distinguish simulation behavior from real-world fog response.

23.11F | Availability score and robustness

Score valid positive reflections N_positive/N_slots; preserve VALID, NO_RETURNS, EMPTY_FRAME, MISSING_FRAME, INVALID_TIMESTAMP and MALFORMED_FRAME. Evaluate real OSI frames and MATLAB comparison.

23.11G/H | Preliminary CPU profile and consolidation

Repeat scoring on fixed frames (600 calls per weather), summarize mean/p95/p99, archive graphs/reports and SHA-256 evidence; no protobuf parse, hardware energy or end-to-end timing included.

23.15 | Unified sensor adapter

Wrap the original LiDAR availability function, verify nine synthetic scenarios, and separately validate zero-rotation local Unreal-to-vehicle OSI conversion. 3D points are not yet integrated into a recorded-OSI end-to-end adapter.

Core mathematical interpretation

Range: a valid simulated time-of-flight corresponds to R = c * TOF / 2 when that sensor field has this round-trip semantics. Availability: R_avail = count(finite positive TOF and strength) / reflection slots. Zero-rotation geometry: P_vehicle,OSI = (x_UE, -y_UE, z_UE) + T_mount,OSI. This convention is source-checked; it is not an independently observed 6-DOF calibration.

Weather	Mean scoring (us)	P95 (us)	P99 (us)
Clear	925.187	967.010	1256.396
Fog100	932.944	965.305	1365.312
Fog20	890.777	926.440	1073.268

Workload timing is Python on the development PC, fixed-frame repeated execution, disk I/O and OSI decoding excluded. No CPU-FPGA energy savings can be derived.

Illustrated figure-by-figure interpretation

Every figure below is embedded from the supplied archive. Captions explain the research question, what the figure can support, and any key interpretation limit. Images are evidence from historical subtasks, not independently recreated measurements.

Task 23.4 - raw range and MATLAB reconstruction

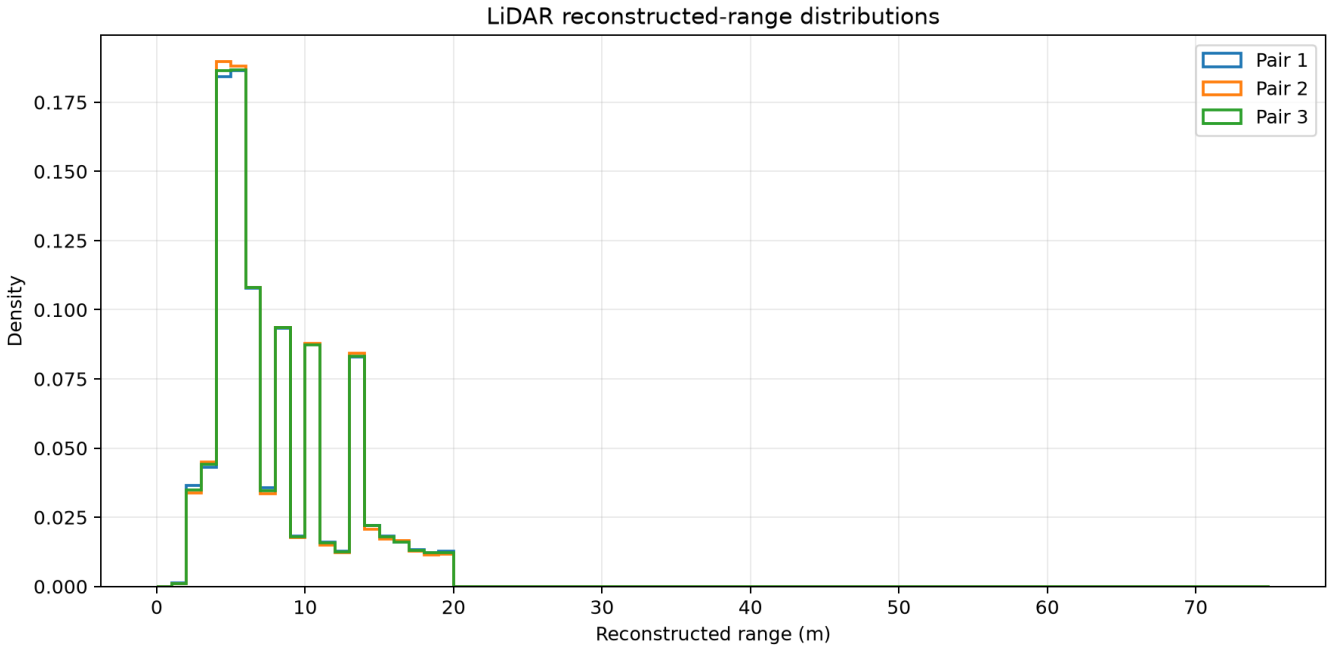


Figure 1. task23_04b_lidar_range_distribution.png - Distribution of valid reconstructed LiDAR ranges; investigate truncation, unexpected peaks or geometry, not target identity.

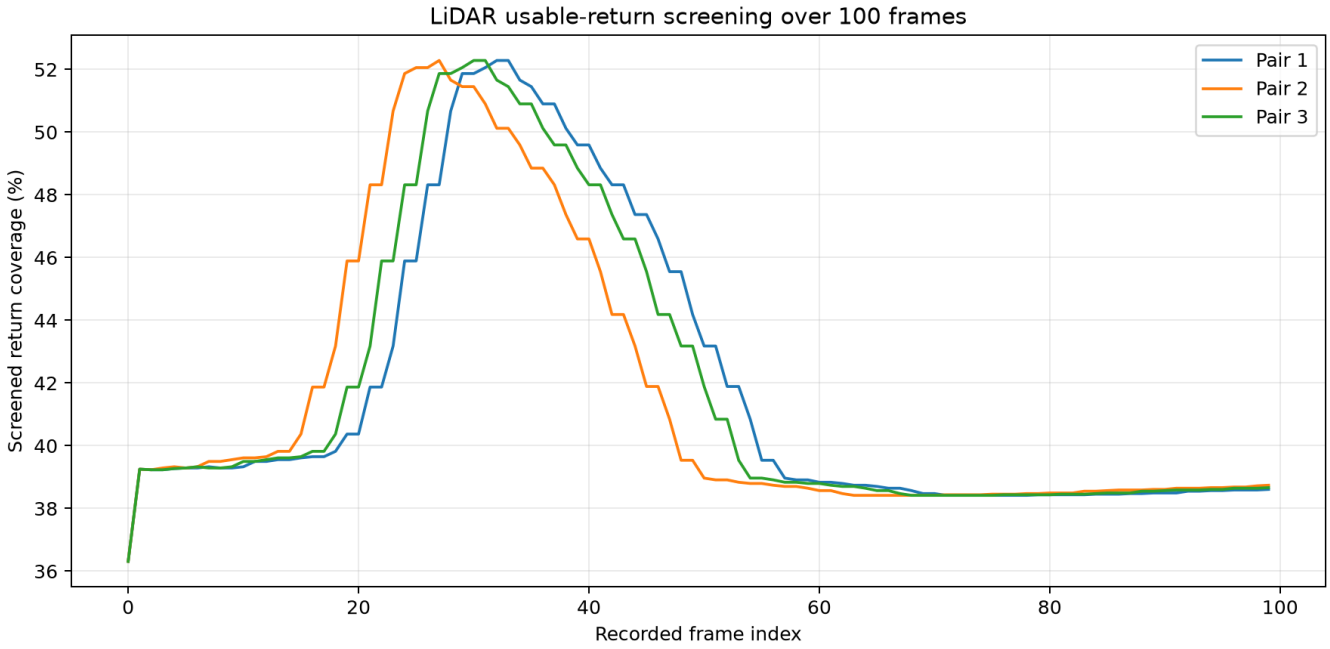


Figure 2. task23_04b_lidar_return_coverage.png - Shows which rays returned usable signals; this is measurement availability, not correct vehicle detections.

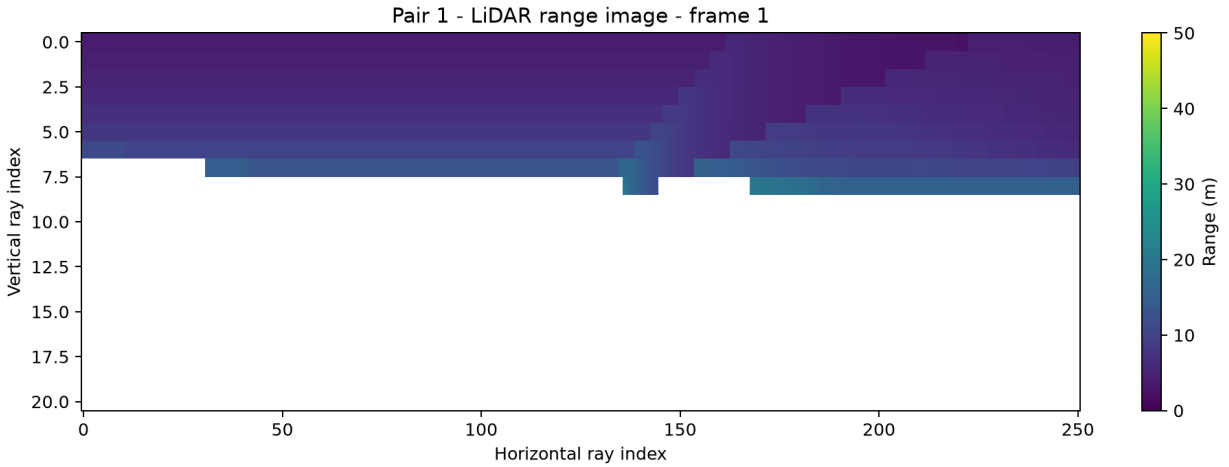


Figure 3. task23_04b_pair1_frame001_range_image.png - A scan indexed by LiDAR ray geometry; patterns are specific to the saved frame and sensor setting.

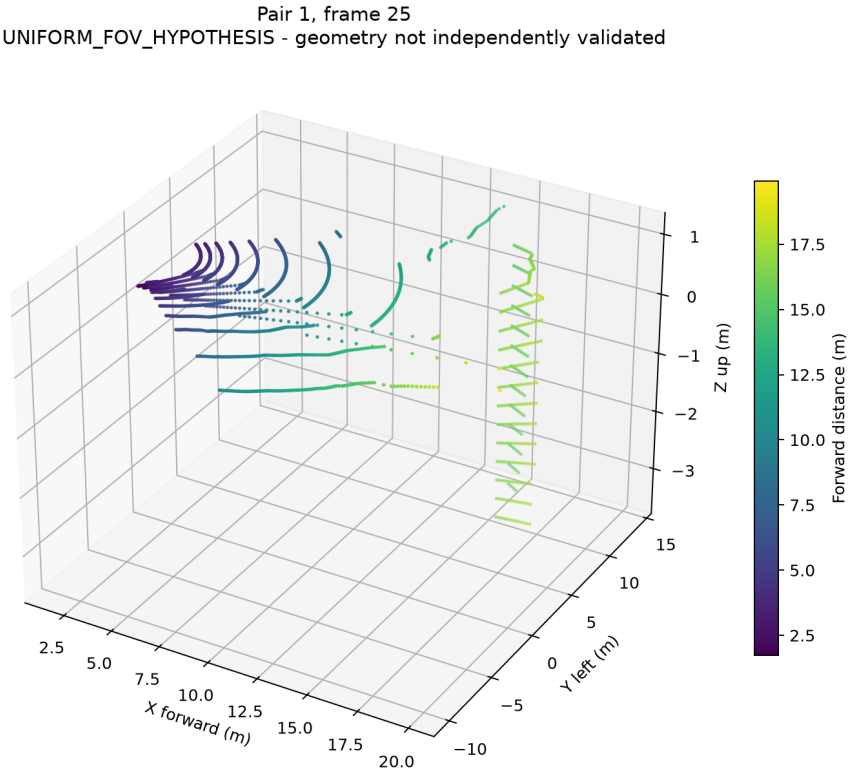


Figure 4. task23_04b_pair1_frame025_point_cloud.png - 3D LiDAR/Radar reconstruction of a sampled static scene; visual spatial coherence is not independently established localization accuracy.

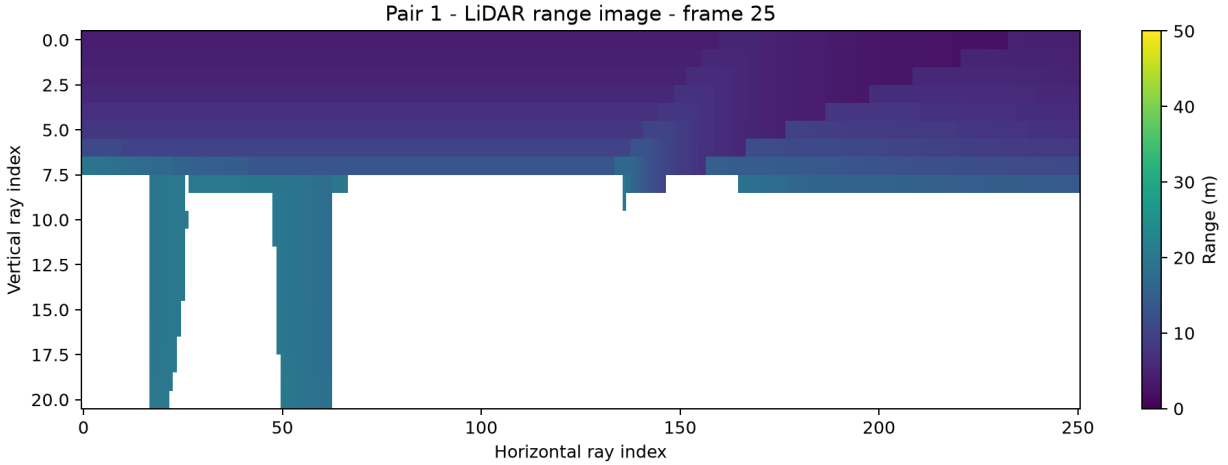


Figure 5. task23_04b_pair1_frame025_range_image.png - A scan indexed by LiDAR ray geometry; patterns are specific to the saved frame and sensor setting.

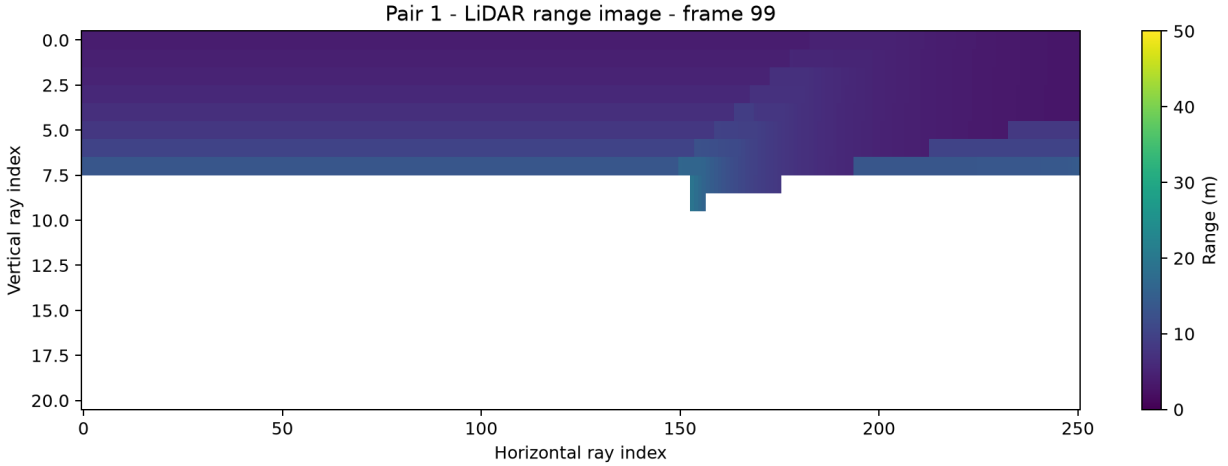


Figure 6. task23_04b_pair1_frame099_range_image.png - A scan indexed by LiDAR ray geometry; patterns are specific to the saved frame and sensor setting.

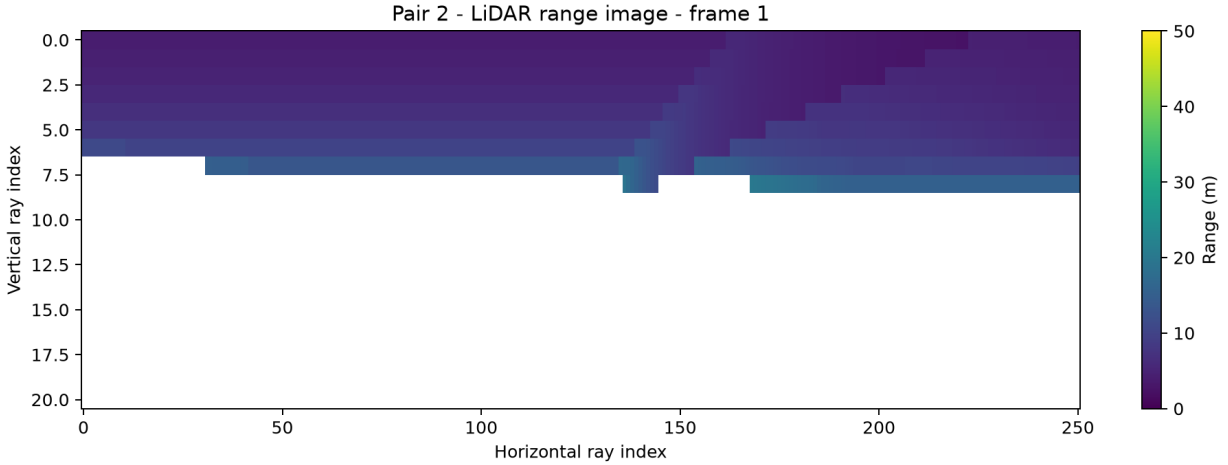


Figure 7. task23_04b_pair2_frame001_range_image.png - A scan indexed by LiDAR ray geometry; patterns are specific to the saved frame and sensor setting.

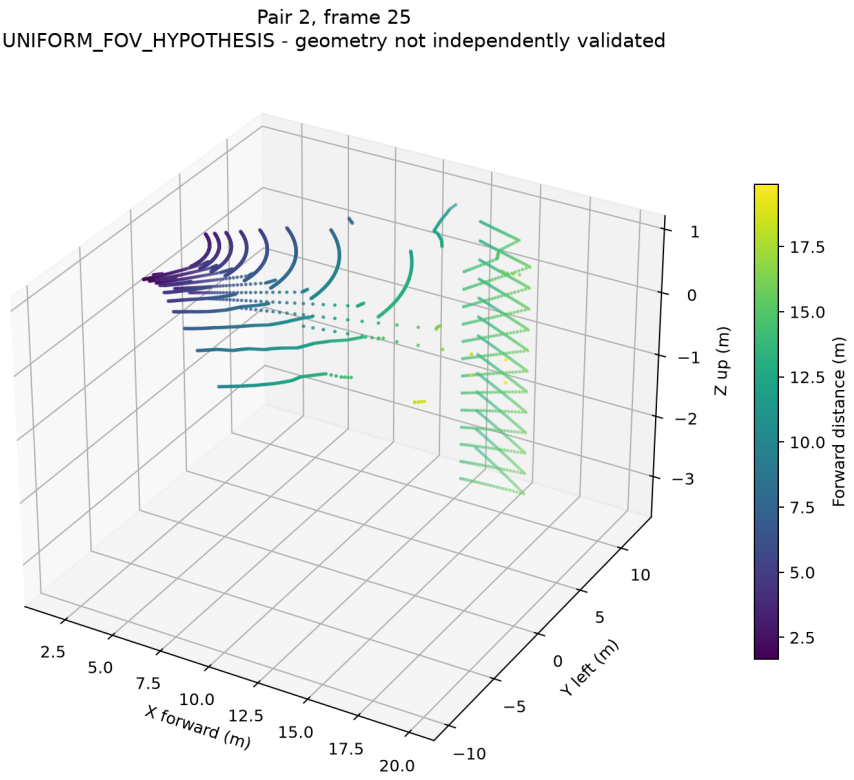


Figure 8. task23_04b_pair2_frame025_point_cloud.png - 3D LiDAR/Radar reconstruction of a sampled static scene; visual spatial coherence is not independently established localization accuracy.

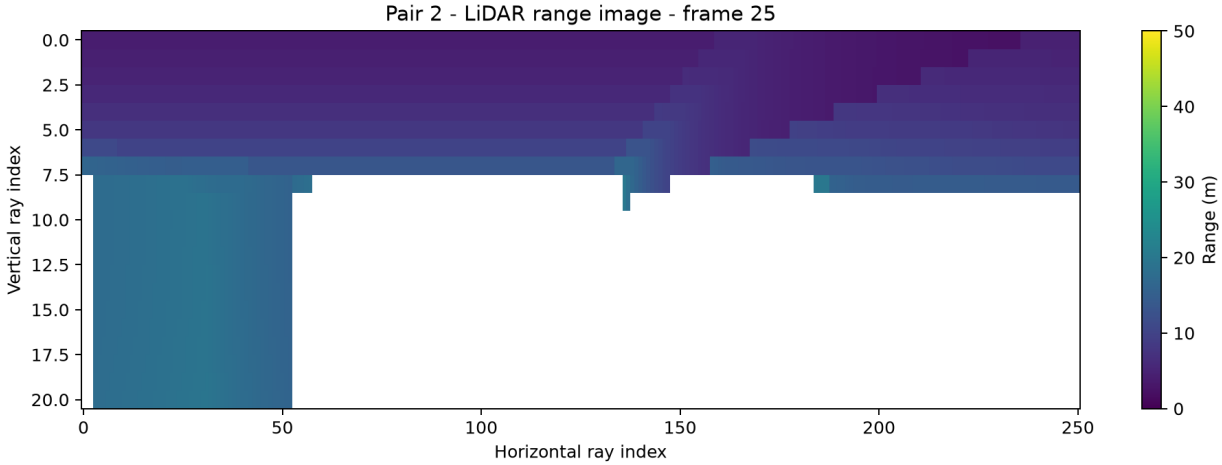


Figure 9. task23_04b_pair2_frame025_range_image.png - A scan indexed by LiDAR ray geometry; patterns are specific to the saved frame and sensor setting.

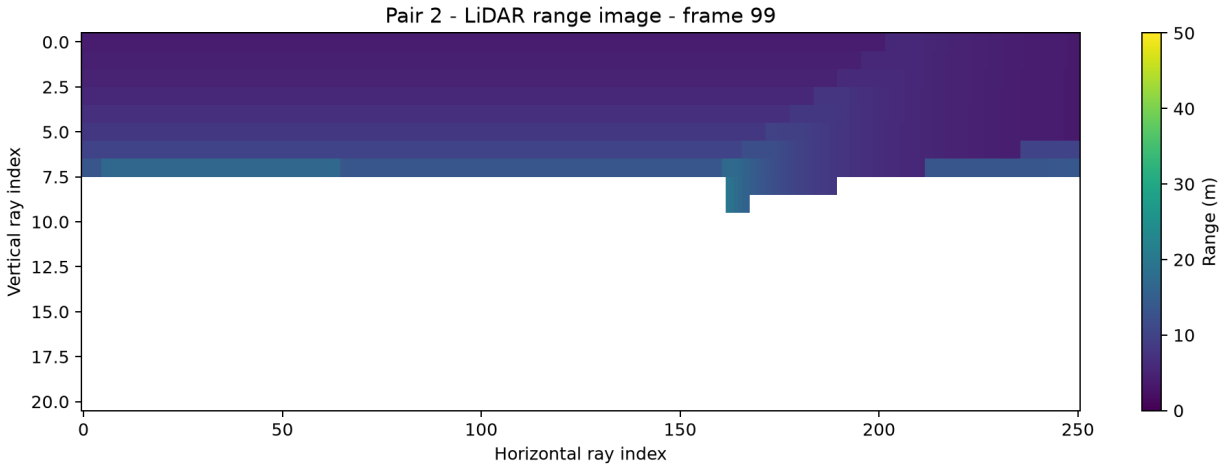


Figure 10. task23_04b_pair2_frame099_range_image.png - A scan indexed by LiDAR ray geometry; patterns are specific to the saved frame and sensor setting.

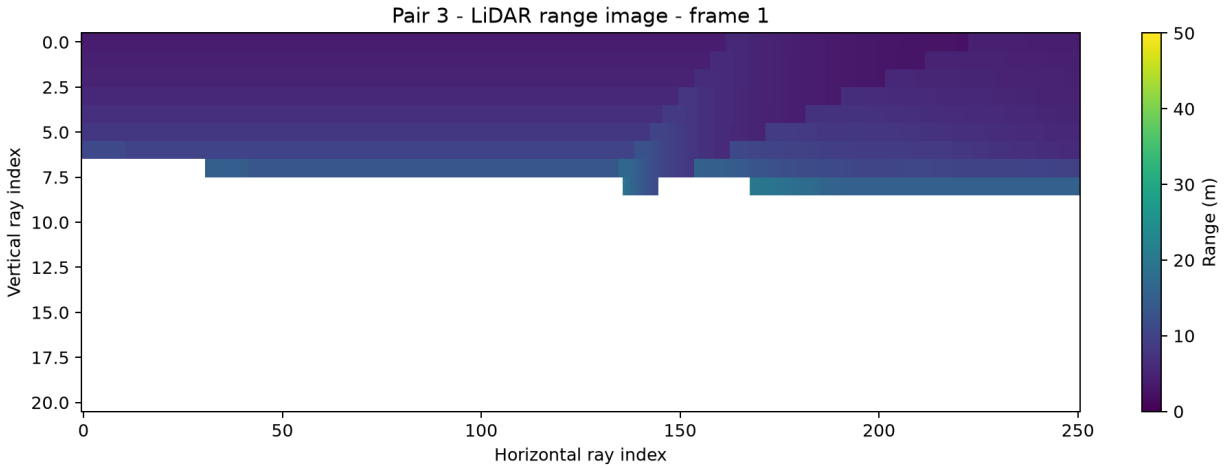


Figure 11. task23_04b_pair3_frame001_range_image.png - A scan indexed by LiDAR ray geometry; patterns are specific to the saved frame and sensor setting.

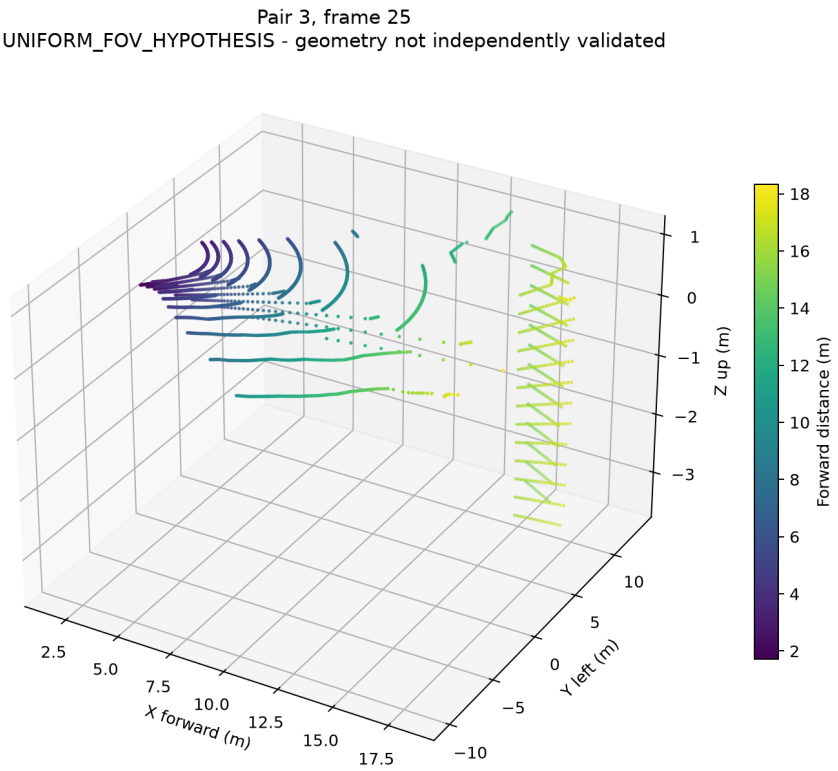


Figure 12. task23_04b_pair3_frame025_point_cloud.png - 3D LiDAR/Radar reconstruction of a sampled static scene; visual spatial coherence is not independently established localization accuracy.

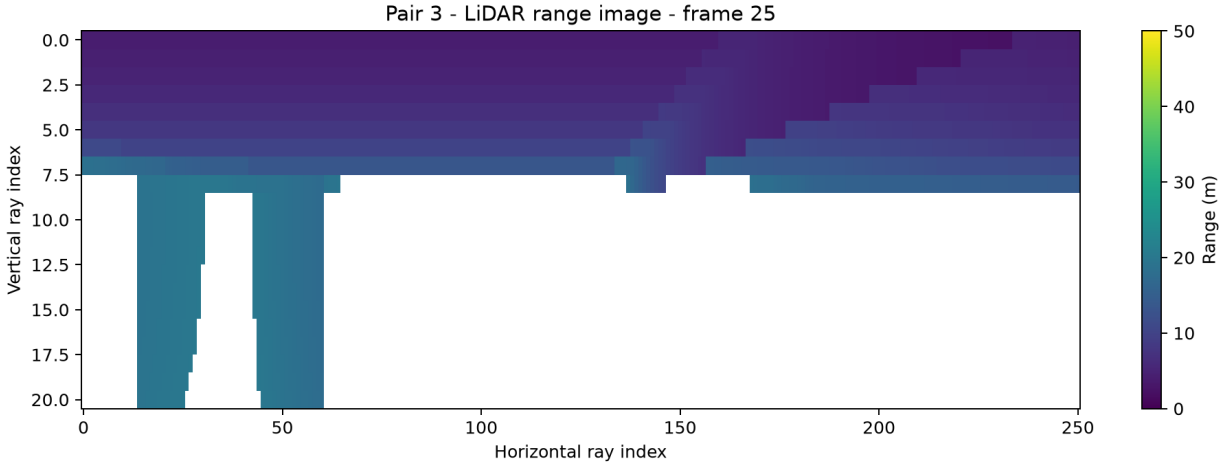


Figure 13. task23_04b_pair3_frame025_range_image.png - A scan indexed by LiDAR ray geometry; patterns are specific to the saved frame and sensor setting.

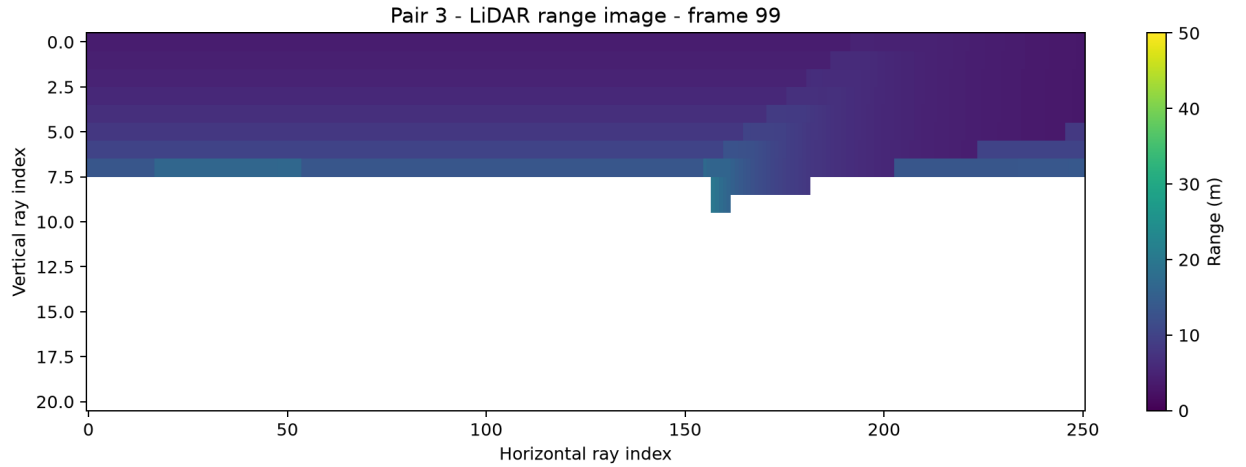


Figure 14. task23_04b_pair3_frame099_range_image.png - A scan indexed by LiDAR ray geometry; patterns are specific to the saved frame and sensor setting.

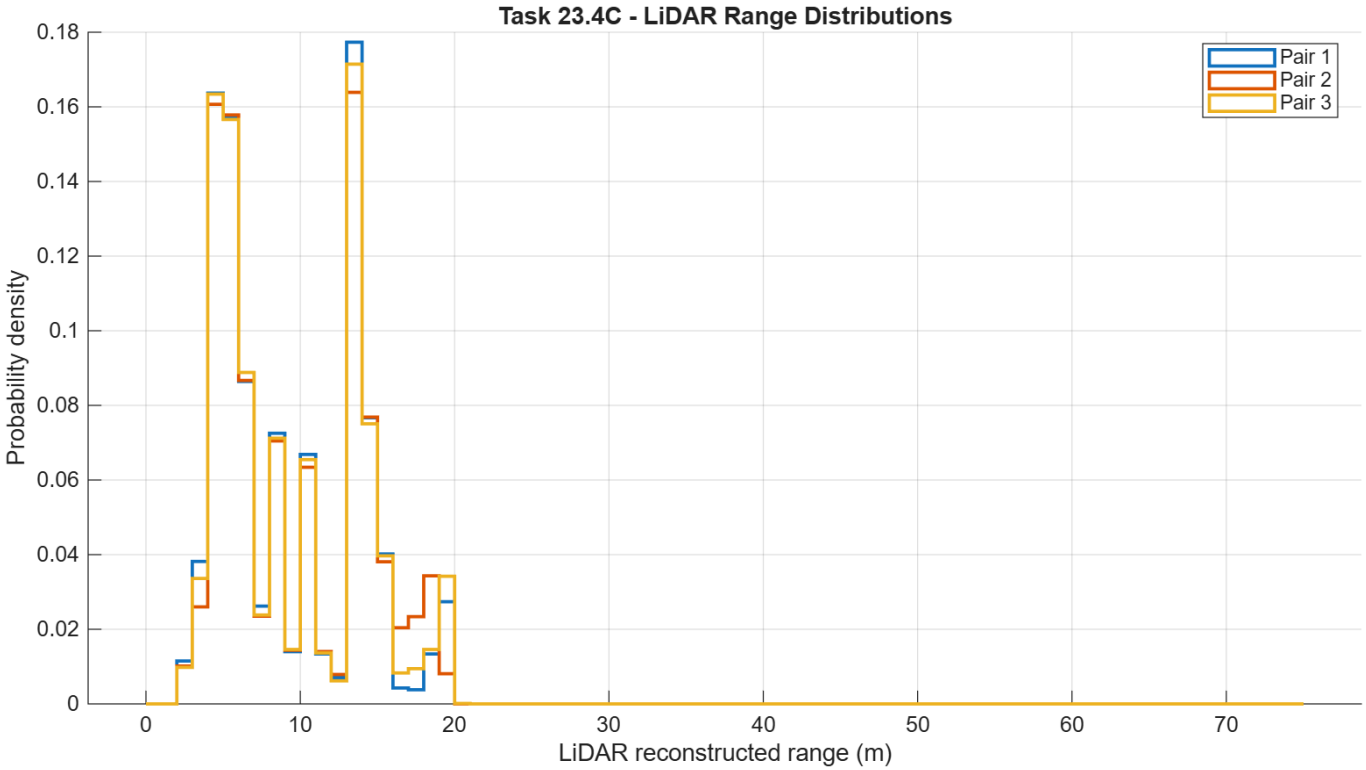


Figure 15. task23_04c2_matlab_range_distributions.png - MATLAB cross-check of LiDAR range histograms from previously exported data.

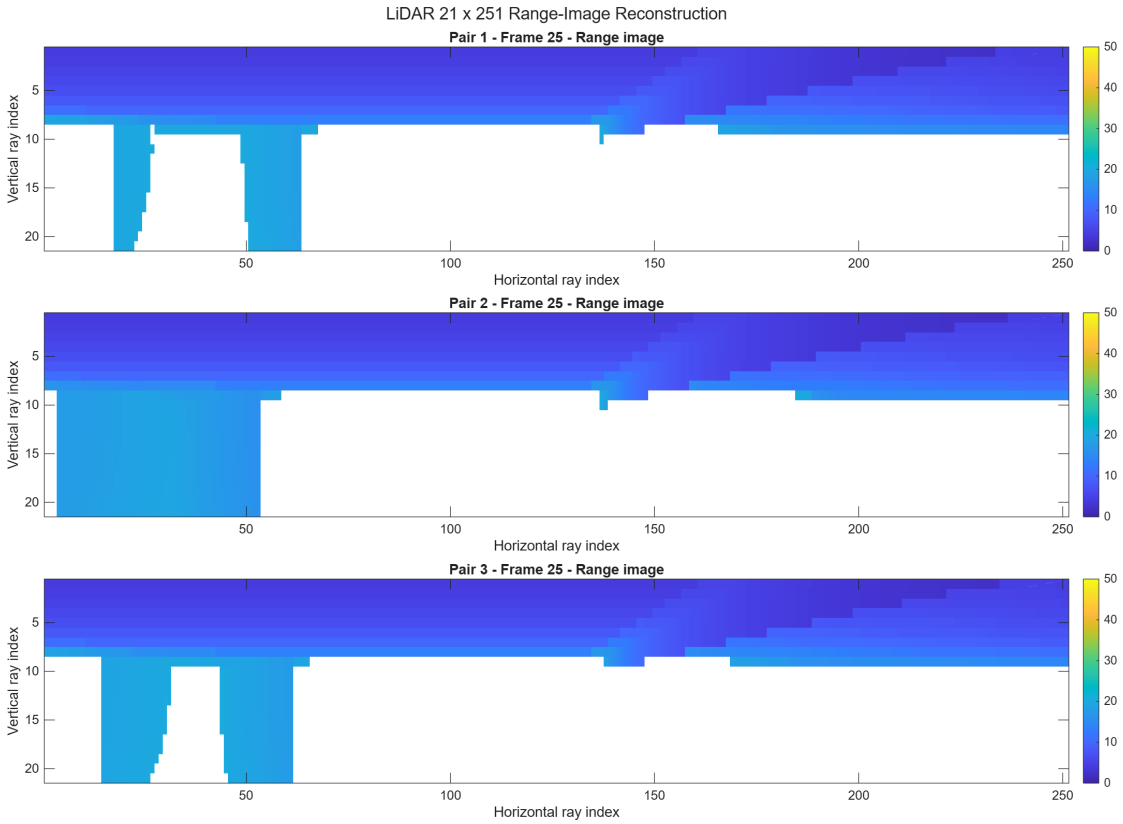


Figure 16. task23_04c2_matlab_range_images.png - MATLAB reconstruction of LiDAR range-image representation; compare scan orientation and missing rays.

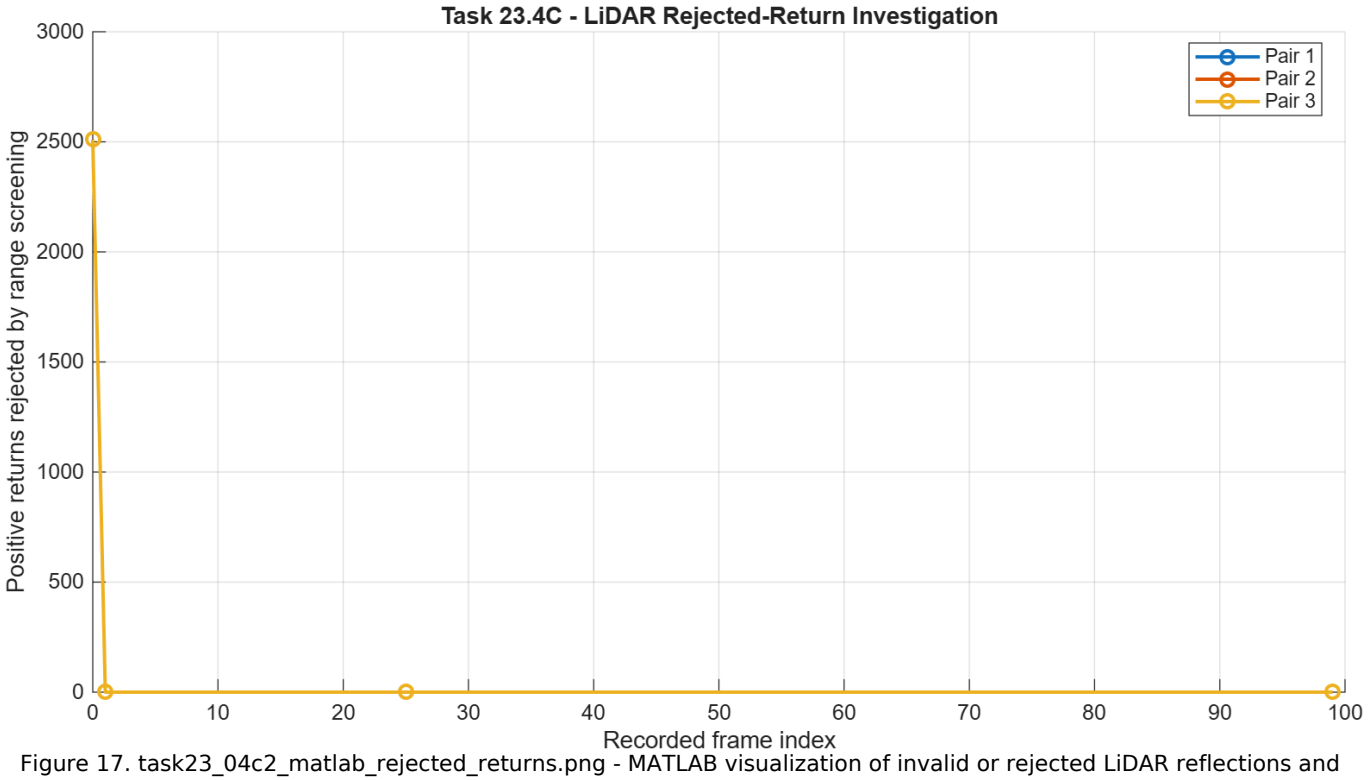


Figure 17. task23_04c2_matlab_rejected_returns.png - MATLAB visualization of invalid or rejected LiDAR reflections and filtering behavior.

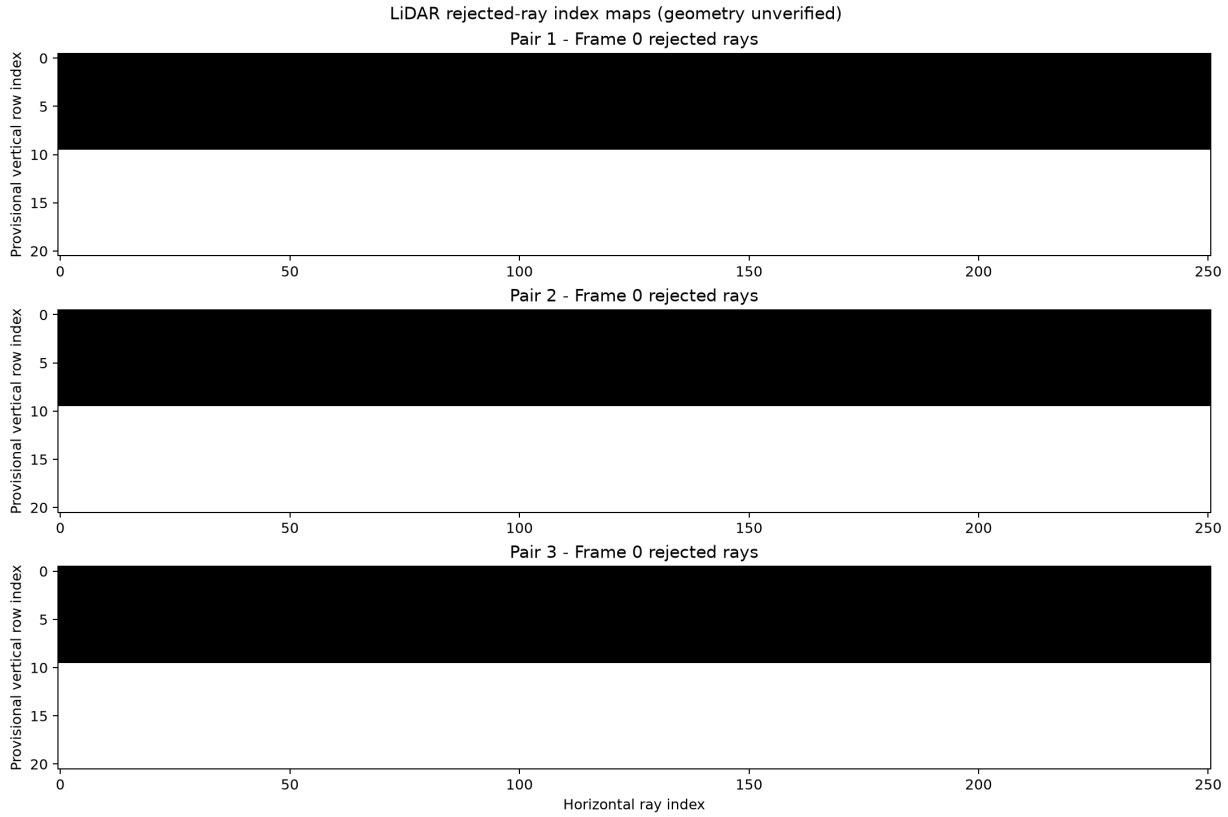


Figure 18. task23_04d1_rejected_ray_maps.png - Spatial map of invalid ray indices; useful for detecting scanning-pattern and indexing effects.

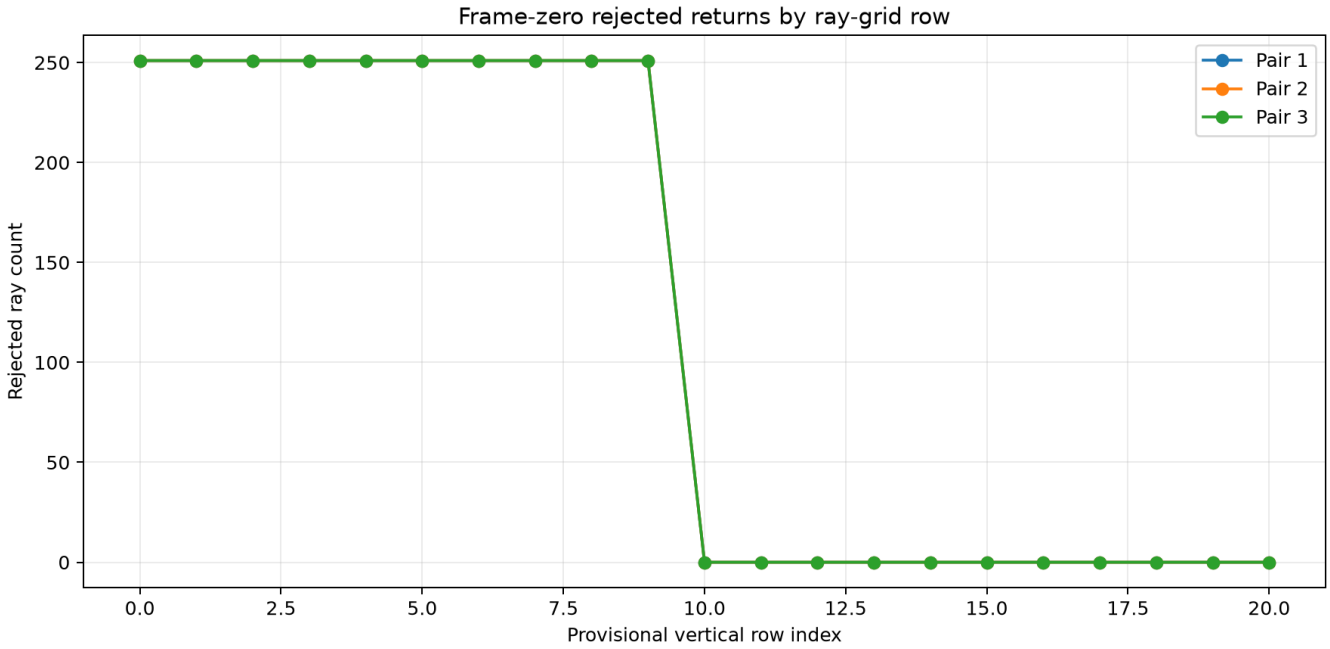


Figure 19. task23_04d1_rejected_row_profile.png - Ray rejection counts by vertical scanning row; checks systematic indexing geometry.

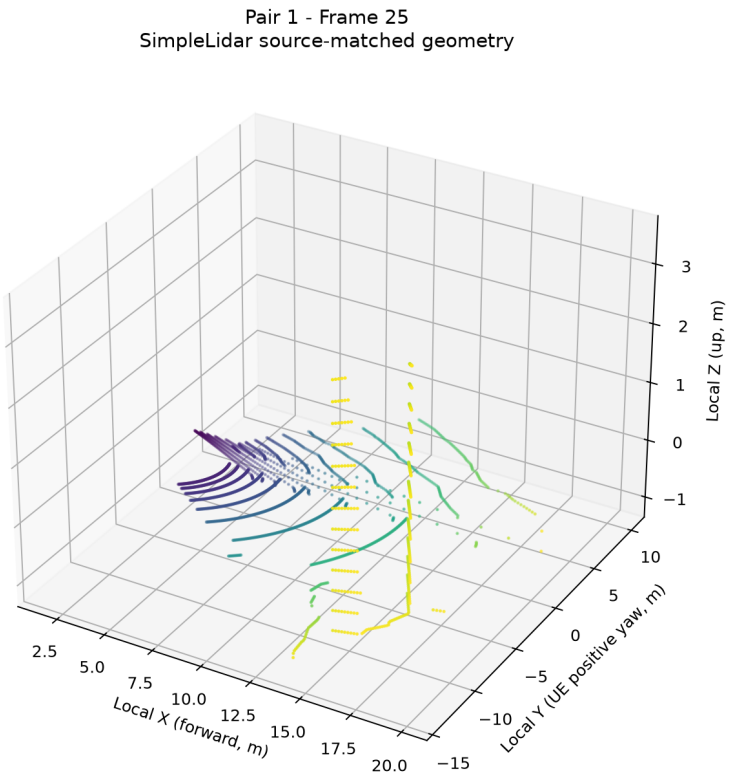


Figure 20. task23_04d4_pair1_corrected_point_cloud.png - Updated point cloud after correcting the source-defined LiDAR ray layout; compare qualitatively to earlier point cloud.

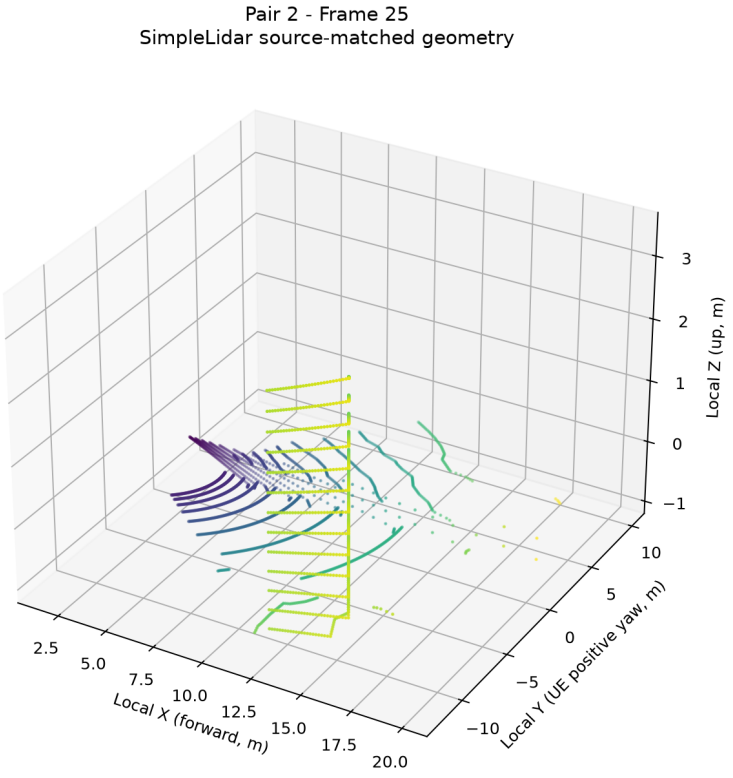


Figure 21. task23_04d4_pair2_corrected_point_cloud.png - Updated point cloud after correcting the source-defined LiDAR ray layout; compare qualitatively to earlier point cloud.

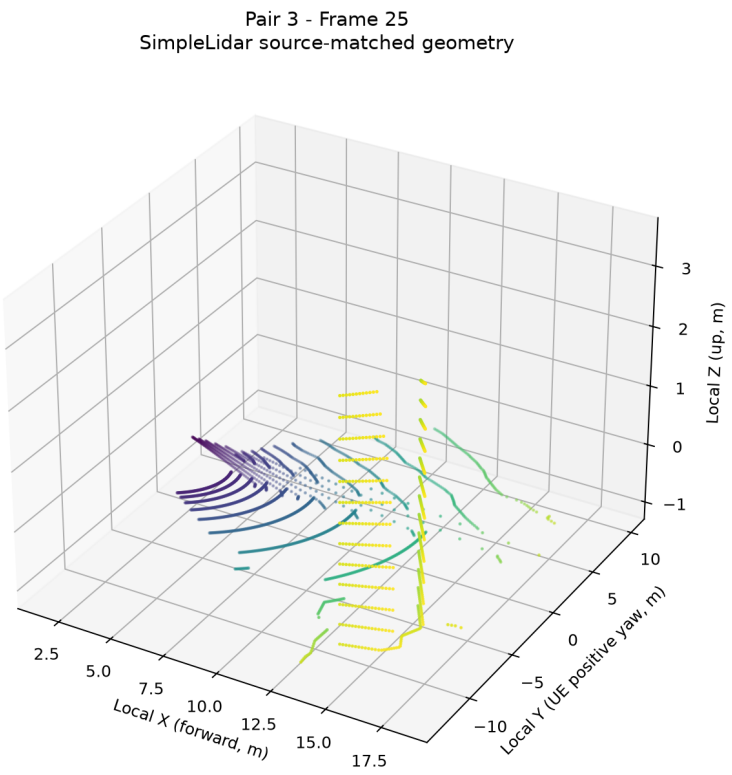


Figure 22. task23_04d4_pair3_corrected_point_cloud.png - Updated point cloud after correcting the source-defined LiDAR ray layout; compare qualitatively to earlier point cloud.

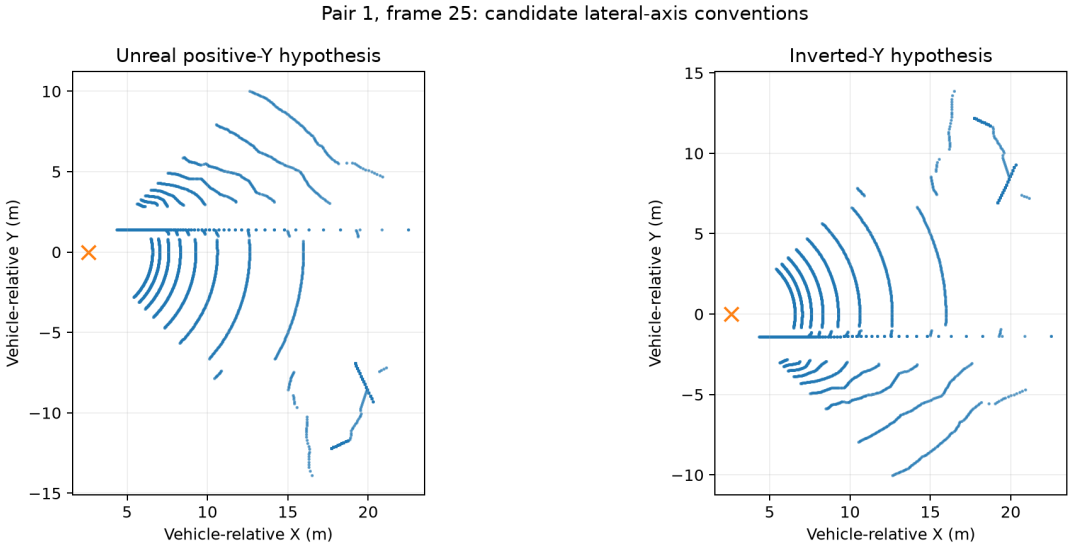


Figure 23. task23_04e3_pair1_vehicle_axis_comparison.png - Compares alternative vehicle-frame coordinates; check sign and origin assumptions before association.

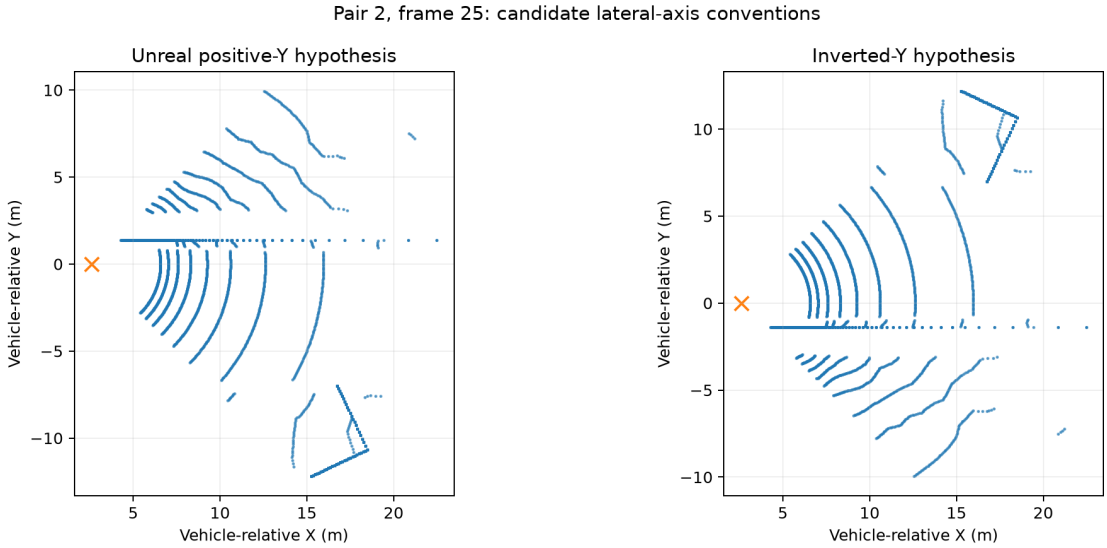


Figure 24. task23_04e3_pair2_vehicle_axis_comparison.png - Compares alternative vehicle-frame coordinates; check sign and origin assumptions before association.

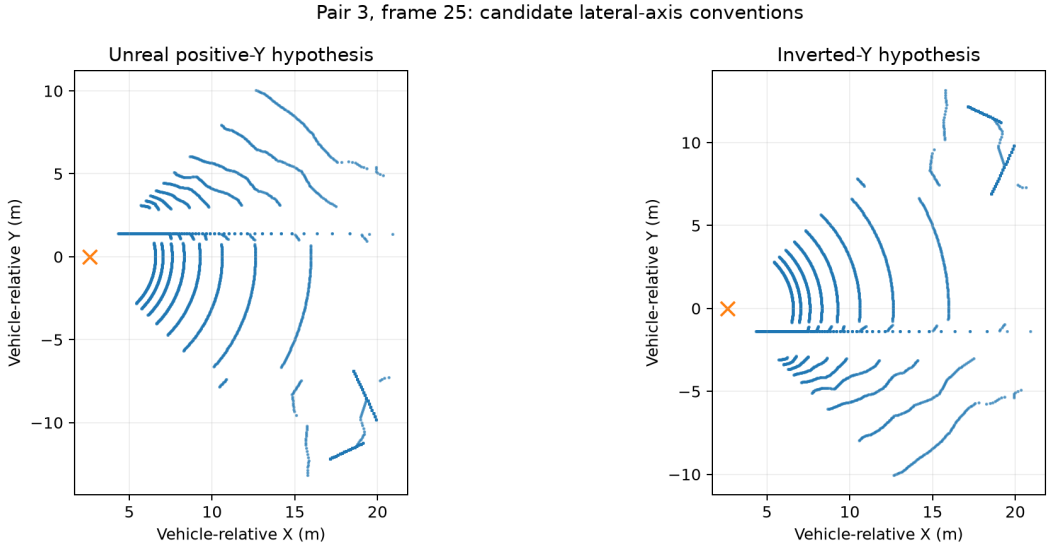


Figure 25. task23_04e3_pair3_vehicle_axis_comparison.png - Compares alternative vehicle-frame coordinates; check sign and origin assumptions before association.

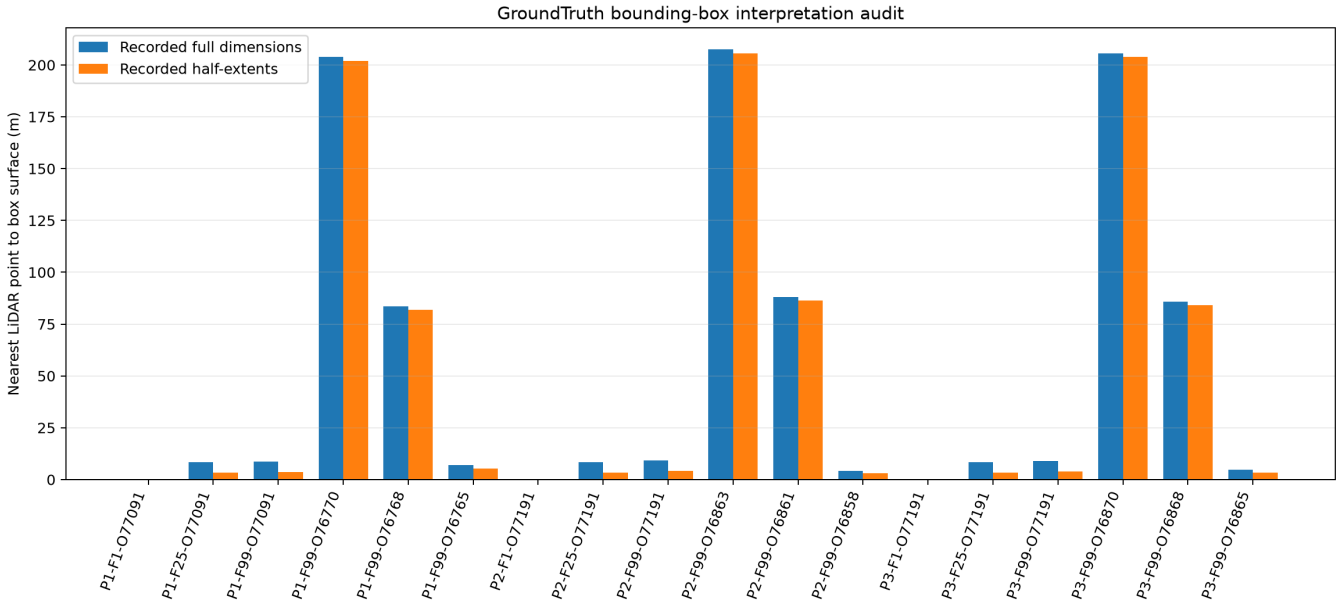


Figure 26. task23_04e5d_bounding_box_comparison.png - Explores geometric bounding-box interpretations; not a verified actor association or object accuracy measure.

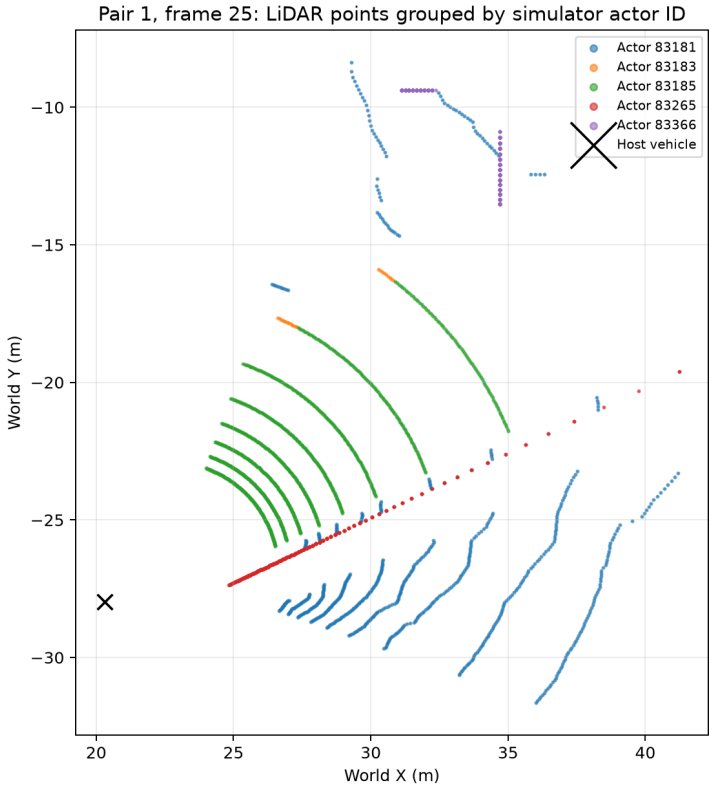


Figure 27. task23_04e5f_actor_id_world_distribution.png - Spatial distribution of actor IDs or ground-truth references; exploratory correspondence diagnostics.

Task 23.11 - weather, reliability and CPU scoring

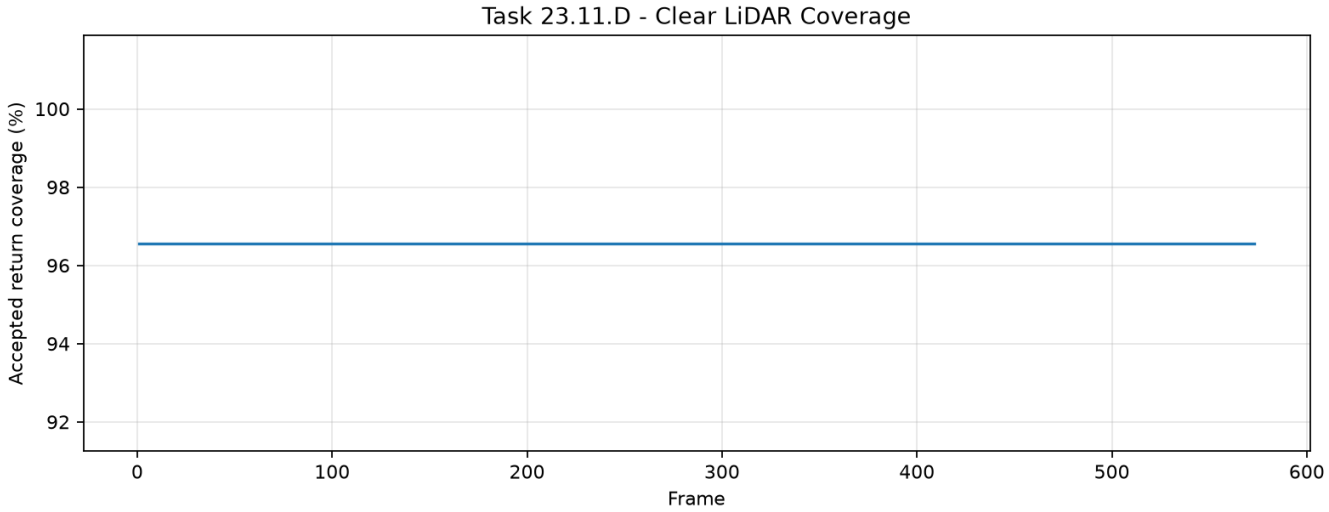


Figure 28. task23_11D_clear_coverage.png - Clear-scene LiDAR returns or coverage baseline for later weather contrasts.

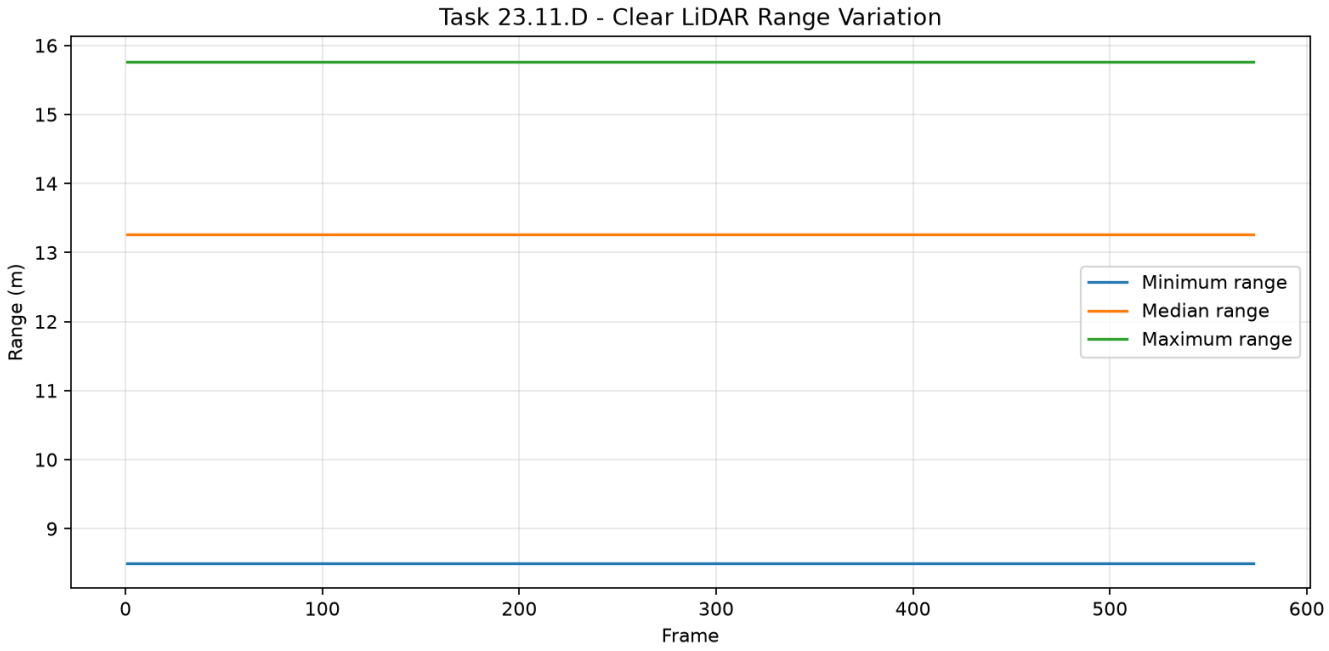


Figure 29. task23_11D_clear_ranges.png - Clear-scene valid-range distribution baseline; useful as control for simulated fog comparisons.

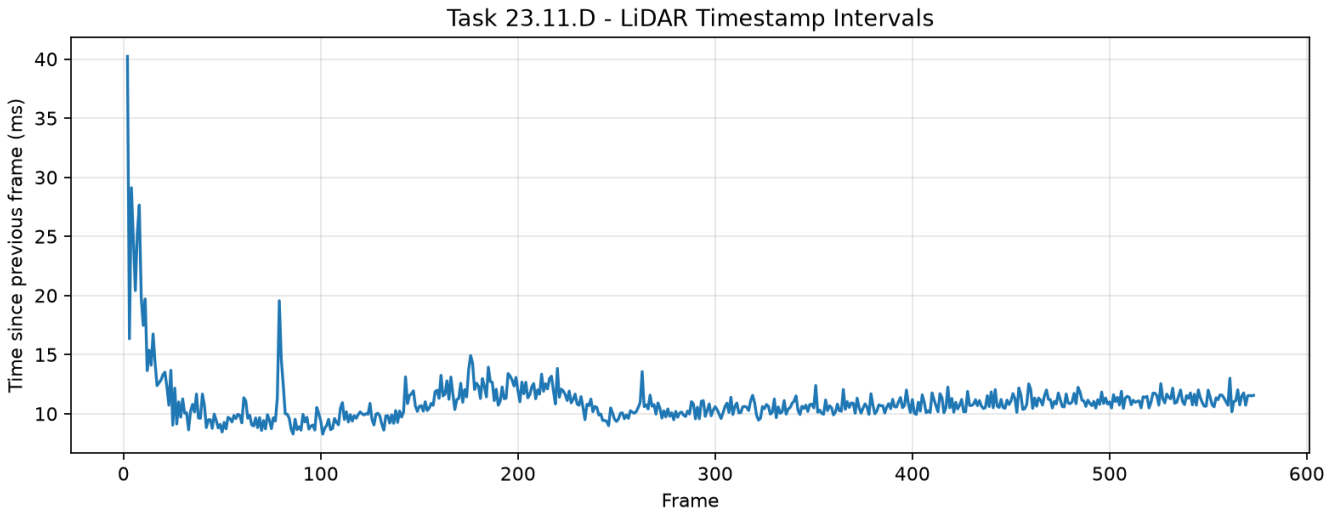


Figure 30. task23_11D_timestamp_intervals.png - Simulation timestamp interval variation; interval is not the relative acquisition-time difference between different sensors.

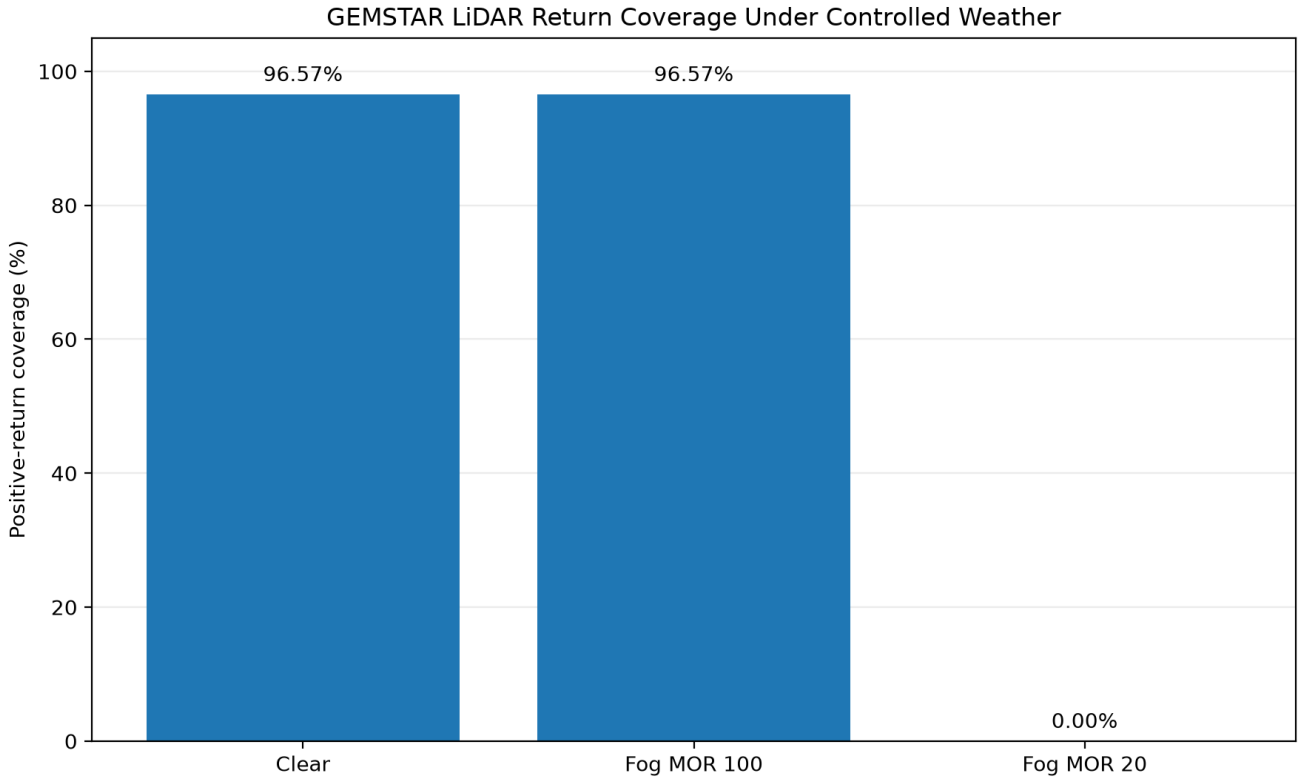


Figure 31. task23_11E_lidar_coverage_by_weather.png - Controlled LiDAR return availability across Clear/Fog100/Fog20; Fog100 matches Clear while Fog20 has no positive returns in the tested static scene.

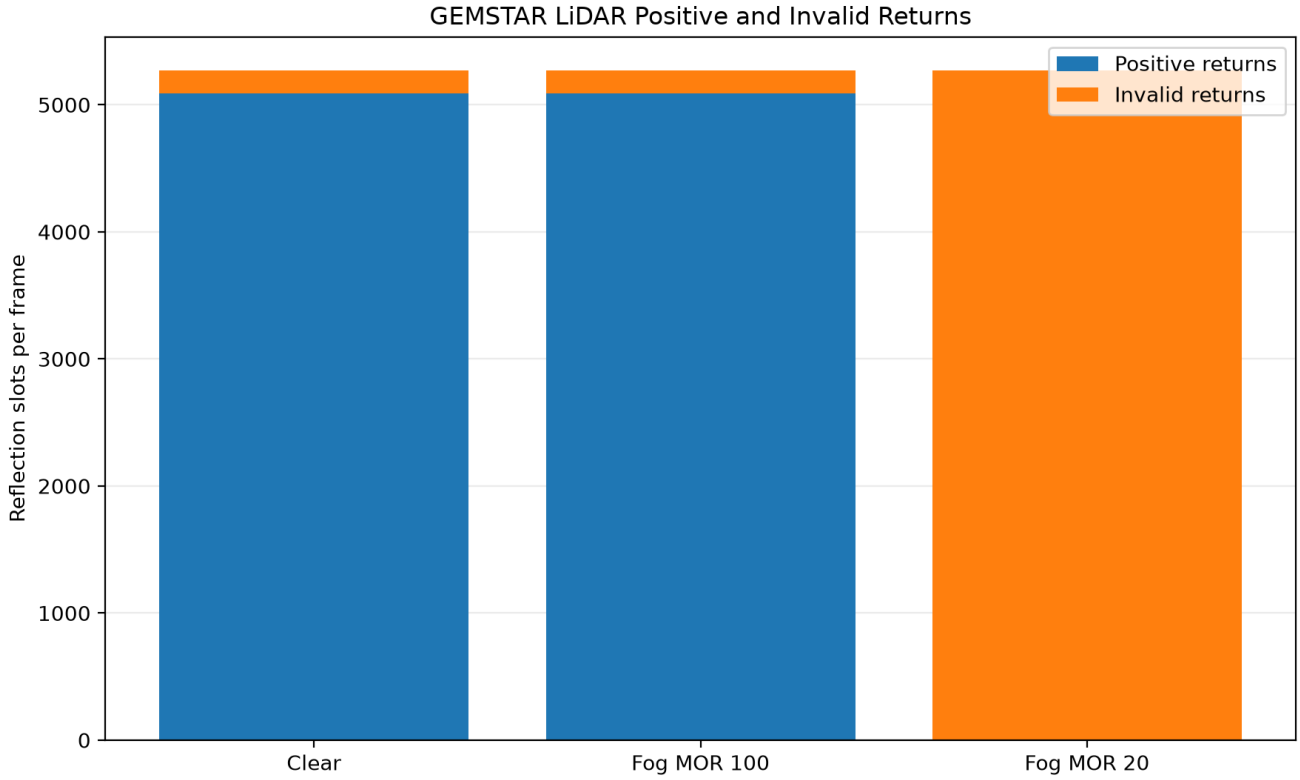


Figure 32. task23_11E_positive_vs_invalid_returns.png - Compares positive and invalid reflection slots; counts do not establish ground-truth detection correctness.

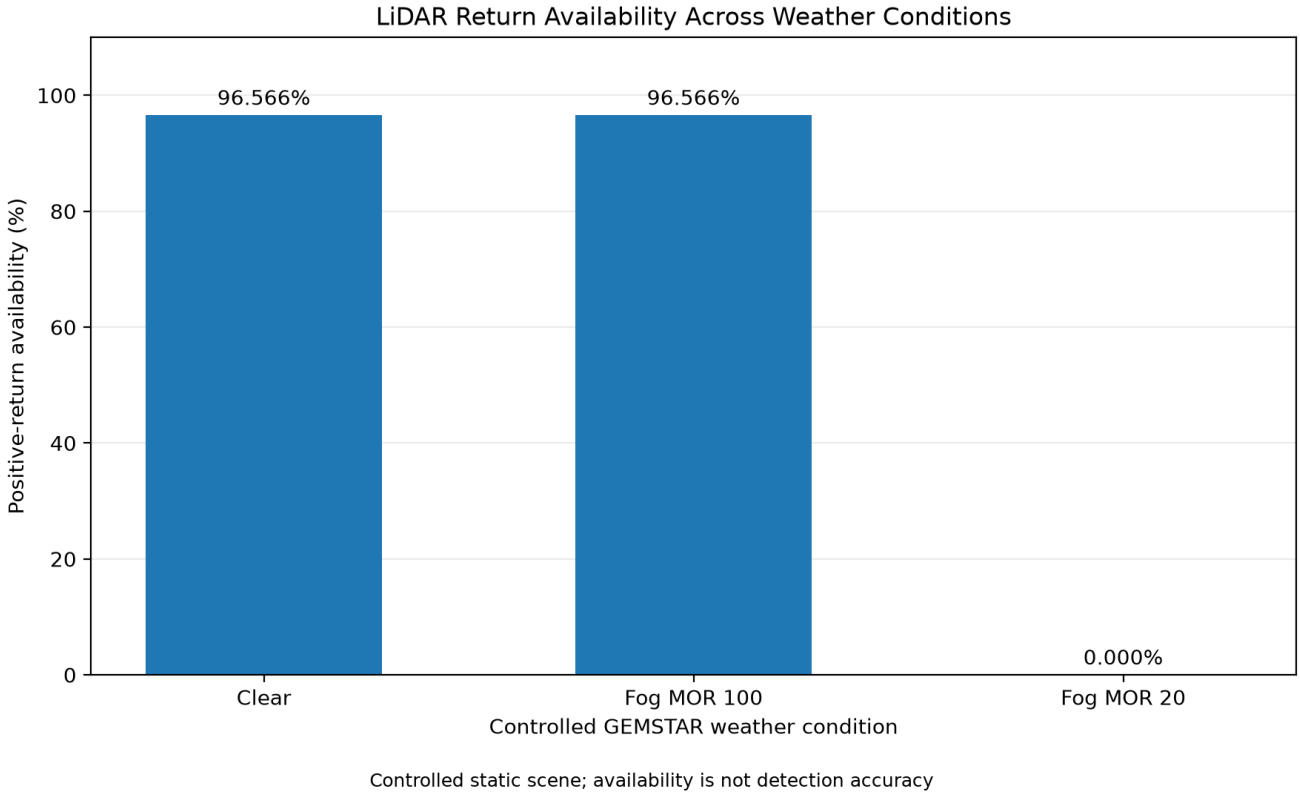


Figure 33. task23_11F_07A_weather_reliability.png - Plots availability ratio for each controlled weather scenario. The label reliability is an availability proxy.

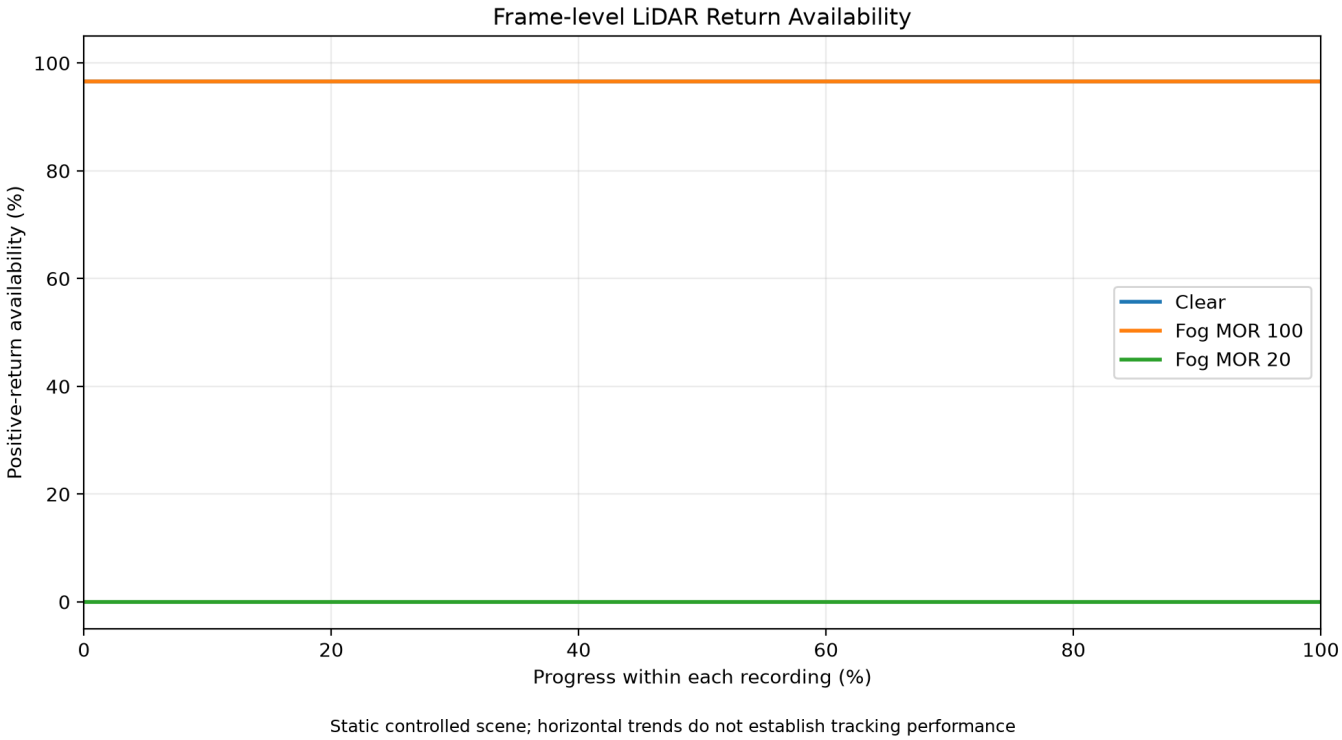


Figure 34. task23_11F_07B_frame_reliability.png - Frame-level availability score; repeated identical static frames are not independent dynamic trials.

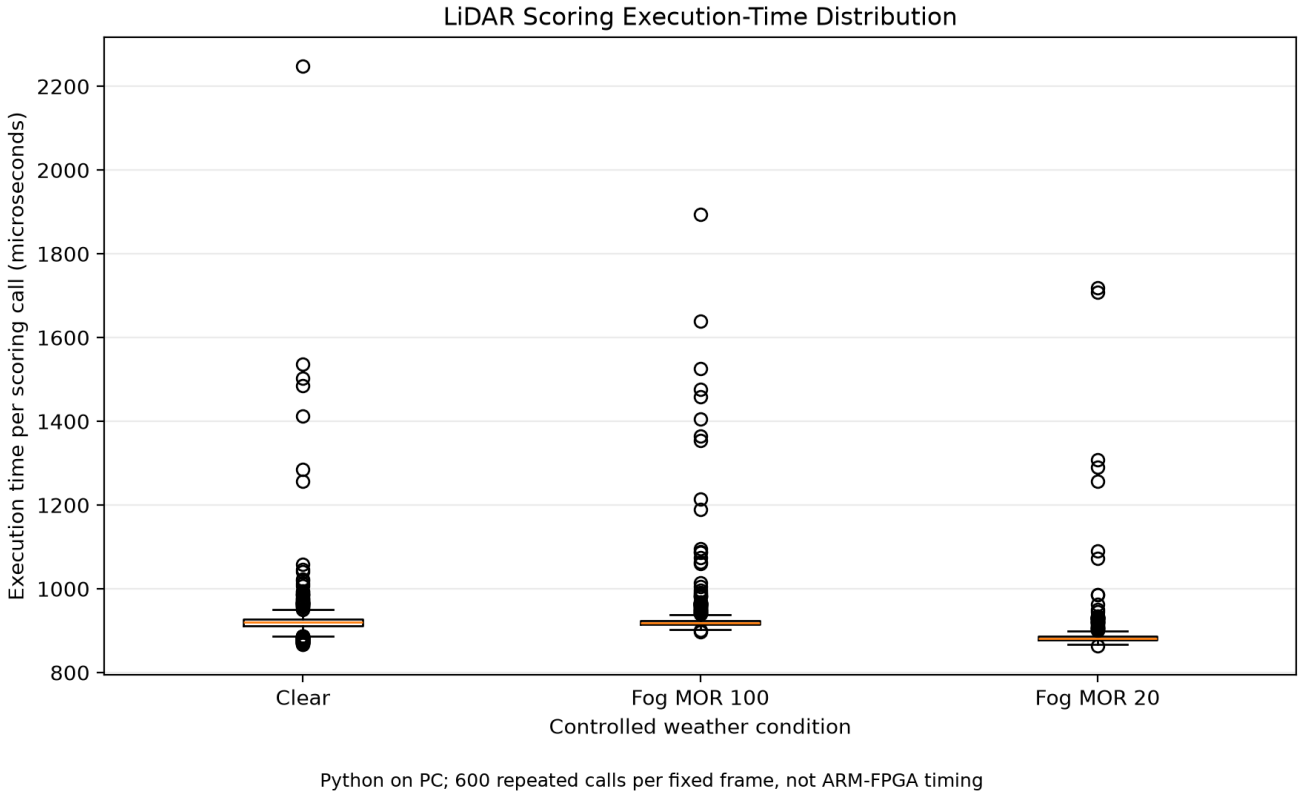


Figure 35. task23_11G_02B_timing_distribution.png - Python desktop LiDAR scoring-call duration; excludes OSI parsing and disk reads and is not a board energy/latency result.

Detailed subtask and code inventory

The table below is organized by archived script name and research objective. A filename identifies work that exists; it is not by itself a claim that every planned experiment or verification was completed.

Task 04a - source scripts (1)

task23_04a_lidar_inspection.py - Inspect recorded or simulated payloads before accepting analysis.

Task 04b - source scripts (1)

task23_04b_lidar_range_reconstruction.py - Reconstruct geometry, range, or Cartesian coordinates.

Task 04c - source scripts (2)

task23_04c1_export_lidar_raw_for_matlab.py - MATLAB cross-tool calculation/plot based on exported research data.

task23_04c2_matlab_lidar_validation.m - MATLAB cross-tool calculation/plot based on exported research data.

Task 04d - source scripts (3)

task23_04d1_lidar_ray_index_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_04d2_lidar_source_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_04d4_correct_lidar_geometry.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 04e - source scripts (10)

task23_04e1_groundtruth_pose_inspection.py - Inspect recorded or simulated payloads before accepting analysis.

task23_04e2_host_pose_recovery.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_04e3_lidar_vehicle_transform.py - Apply or check reference-frame transformations.

task23_04e4d_finalize_vehicle_coordinates.py - Audit or finalize coordinate frame and geometry conventions.

task23_04e5a_world_transform.py - Apply or check reference-frame transformations.

task23_04e5b_groundtruth_geometry_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_04e5c_host_object_association.py - Explore correspondence candidates and rejection or identity assumptions.

task23_04e5d_bounding_box_consistency.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_04e5e_lidar_object_id_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_04e5f_actor_spatial_consistency.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 08 - source scripts (11)

task23_08_clear_set_map.py - Modify or prepare experimental map (historical operation; do not rerun).

task23_08_clear_verify_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_08_compare_clear_fog.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_08_displace_target.py - Controlled change to target geometry for calibration.

task23_08_finalize_lidar.py - Consolidate evidence and create a research-ready result.

task23_08_inspect_lidar_osi.py - Inspect recorded or simulated payloads before accepting analysis.

task23_08_inspect_world_config.py - Inspect recorded or simulated payloads before accepting analysis.

task23_08_set_map_config.py - Modify or prepare experimental map (historical operation; do not rerun).

task23_08_validate_displacement.py - Test or numerical validation of a prior dataset/implementation.

task23_08_verify_displacement.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_08_verify_map_config.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 10 - source scripts (5)

task23_10_inspect_map_reference.py - Inspect recorded or simulated payloads before accepting analysis.

task23_10_set_fog_map.py - Prepare controlled weather configuration (historical operation).

task23_10_set_weather_map.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_10_verify_fog_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_10_verify_weather_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 11B - source scripts (5)

task23_11B_displaced_clear_sensorview.py - Controlled change to target geometry for calibration.

task23_11B_full_frame_integrity.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11B_inspect_actual_timestamps.py - Inspect recorded or simulated payloads before accepting analysis.

task23_11B_inspect_timestamp_schema.py - Inspect recorded or simulated payloads before accepting analysis.

task23_11B_preliminary_fog_sensorview.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 11C - source scripts (3)

task23_11C_inspect_npz.py - Inspect recorded or simulated payloads before accepting analysis.

task23_11C_validate_lidar_masks.py - Test or numerical validation of a prior dataset/implementation.

task23_11C_validate_npz.py - Test or numerical validation of a prior dataset/implementation.

Task 11D - source scripts (4)

task23_11D_baseline_graphs.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11D_baseline_stability.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11D_clear_baseline.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11D_compare_raw_frames.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 11E - source scripts (12)

task23_11E_clear_distance_thresholds.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11E_compare_first_reflections.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11E_full_clear_variability.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11E_generate_weather_graphs.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11E_sample_clear_osi.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11E_sample_clear_osi_fixed.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11E_set_clear_map.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11E_set_fog100_map.py - Prepare controlled weather configuration (historical operation).

task23_11E_set_fog20_map.py - Prepare controlled weather configuration (historical operation).

task23_11E_verify_clear_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_11E_verify_fog100_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_11E_verify_fog20_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 11F - source scripts (11)

task23_11F_03B_test_reliability.py - Calculate positive-return availability or handle invalid inputs.

task23_11F_04A_verify_real_osi.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_11F_04B_full_reliability.py - Calculate positive-return availability or handle invalid inputs.

task23_11F_05A_validate_reliability_csv.py - Test or numerical validation of a prior dataset/implementation.

task23_11F_05B_weather_score_validation.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11F_06A_abnormal_reflection_tests.py - Run defined behavior or invalid-input checks.

task23_11F_06B2_malformed_assertions.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11F_06B_missing_malformed_tests.py - Run defined behavior or invalid-input checks.

task23_11F_07A_weather_visualization.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11F_07B_frame_reliability_graph.py - Calculate positive-return availability or handle invalid inputs.

task23_11F_lidar_reliability.py - Calculate positive-return availability or handle invalid inputs.

Task 11G - source scripts (3)

task23_11G_01_preliminary_profile.py - Development-PC compute workload diagnostic, not energy measurement.

task23_11G_02A_timing_distribution.py - Evaluate intervals or processing durations under stated sampling conditions.

task23_11G_02B_timing_analysis.py - Evaluate intervals or processing durations under stated sampling conditions.

Task 15 - source scripts (2)

task23_15_coordinate_adapter.py - Wrap validated sensor-specific operation into common interface.

task23_15_lidar_adapter.py - Wrap validated sensor-specific operation into common interface.

CSV evidence and reproducibility register

CSV file	Rows	Example fields
task23_04a_lidar_validity_audit.csv	28	PairID, Frame, LiDAREntries...
task23_04b_lidar_frame_range_metrics.csv	400	PairID, Frame, RaySlots...
task23_04b_lidar_recording_summary.csv	4	PairID, Frames, RaySlots...

CSV file	Rows	Example fields
task23_04c2_matlab_scan_validation.csv	12	PairID, Frame, RaySlots...
task23_04d1_ray_index_summary.csv	12	PairID, Frame, PositiveReturns...
task23_04d1_vertical_row_audit.csv	252	PairID, Frame, VerticalRowIndex...
task23_04d2_source_file_ranking.csv	55	File, Score, PrimaryMatches...
task23_04d2_source_matches.csv	1283	File, Line, Text...
task23_04d4_geometry_correction_summary.csv	9	PairID, Frame, Points...
task23_04e1_host_pose_summary.csv	9	PairID, Frame, SVTimestampNs...
task23_04e2_host_pose_recovery.csv	9	PairID, Frame, SVTimestampNs...
task23_04e3_vehicle_transform_validation.csv	9	PairID, Frame, Points...
task23_04e4d_source_verified_coordinates.csv	9	PairID, Frame, Points...
task23_04e5a_world_transform_validation.csv	9	PairID, Frame, Points...
task23_04e5b_groundtruth_frame_summary.csv	9	PairID, Frame, LiDARPoints...
task23_04e5b_groundtruth_object_audit.csv	27	PairID, Frame, Category...
task23_04e5c_host_association_summary.csv	9	PairID, Frame, GroundTruthObjects...
task23_04e5c_object_association.csv	27	PairID, Frame, ObjectID...
task23_04e5d_box_interpretation_comparison.csv	18	PairID, Frame, ObjectID...
task23_04e5d_box_surface_metrics.csv	36	PairID, Frame, ObjectID...

CSV file	Rows	Example fields
task23_04e5e_object_id_frame_summary.csv	9	PairID, Frame, ValidLiDARPoints...
task23_04e5e_reflection_object_id_counts.csv	39	PairID, Frame, ReflectionObjectID...
task23_04e5f_actor_recurrence.csv	5	ActorID, ObservationCount, RecordingPairs...
task23_04e5f_actor_spatial_groups.csv	39	PairID, Frame, ActorID...
task23_04e5f_frame1_pair_consistency.csv	3	PairA, PairB, Frame...
task23_08_clear_fog_comparison.csv	10	frame, total_rays, clear_valid...
task23_08_lidar_displacement_rays.csv	5271	ray_index, baseline_m, displaced_m...
task23_10_clear_fog_runtime_results.csv	2	Weather, ControllerSpawned, ManagerRegistered...
task23_11A_source_integrity_audit.csv	7	EvidenceType, FileExists, HashMatch...
task23_11B_displaced_clear_sensorview_integrity.csv	604	Frame, TimestampNS, TimestampPresent...

CSV file	Rows	Example fields
task23_11B_frame_integrity.csv	573	Frame, TimestampNS, TimestampPresent...
task23_11B_preliminary_fog_sensorview_integrity.csv	336	Frame, TimestampNS, TimestampPresent...
task23_11C_npz_integrity.csv	18	Folder, File, ValidRays...
task23_11D_clear_lidar_frame_metrics.csv	573	Frame, TimestampNS, IntervalMS...
task23_11E_three_weather_comparison.csv	3	Weather, MOR_m, TotalFrames...
task23_11F_04B_frame_reliability.csv	40802	weather, frame_index, timestamp_seconds...
task23_11F_04B_reliability_summary.csv	3	weather, frames, positive_returns_per_frame...
task23_11F_07A_matlab_weather_reliability.csv	3	weather_code, frame_count, mean_availability...
task23_11G_01_preliminary_timing.csv	3	weather, repetitions, reflection_slots...
task23_11G_02A_raw_timing_distribution.csv	1800	round, weather, repetition...

CSV file	Rows	Example fields
task23_11G_02B_timing_statistics.csv	3	weather, samples, mean_us...

Missing figures, why they matter, and how to create them

Figure / experiment	Why it is needed	Source / feasibility
Three-weather point-cloud panel from matched sensor poses	Show if visibility/return geometry changes between actual static weather samples, not just counts	Generate only after confirming point-cloud samples have equivalent mount/scene and correct labels.
Multi-frame/dynamic reliability experiment	Current scene patterns are repeated; true reliability needs moving targets, occlusion and GT	Task 24/25, new controlled dynamic runs required.
Ground-truth matched target error distribution	Separates point proximity from true target correspondences	Requires known IDs/geometry and validated transforms.
ARM versus FPGA energy/latency plots	Fundamental future thesis optimization criterion	Later hardware tasks; no board measurements yet.

Scientific interpretation and handover

The checked results are a reproducible software and simulator preprocessing baseline. They do not establish real-world accuracy, dynamic multi-object perception, CPU-versus-FPGA energy improvement, or measured deadlines. Those claims require separate matched scenarios, explicit ground-truth evaluation, and physical hardware benchmarking.

All embedded plots/images are original archived figures except where a figure is explicitly described as a newly synthesized summary. The archived source file name accompanies each caption. Raw multi-gigabyte OSI sources are not embedded and were not reprocessed for these PDFs.